



Gun Detection in Schools

ENSURING SCHOOL SAFETY
WITH GUN DETECTION
TECHNOLOGY

A female police officer in a light grey uniform and dark cap with a star emblem is holding a black walkie-talkie to her mouth. She is looking off to the side with a focused expression. The background is a blurred school building with multiple windows and a green lawn.

Agenda

**THE CHALLENGE OF SCHOOL SAFETY
GUN DETECTION SYSTEMS AS A SOLUTION**

Rising Concerns for School Safety



School Shootings

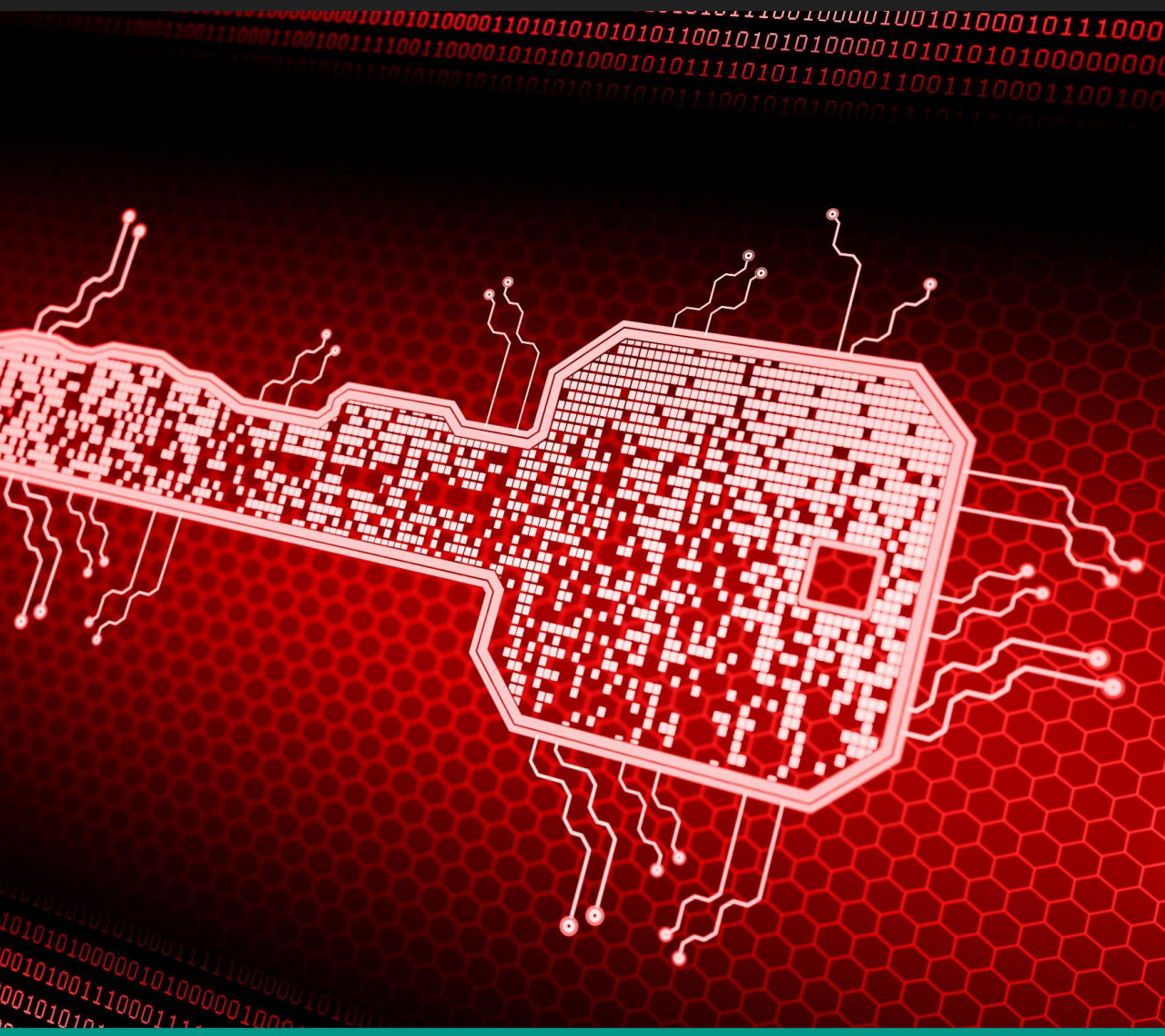
School shootings have become a tragic reality of our times with the school shootings reported in the US, raising concerns about school safety and security.

Importance of School Safety

The safety of students and staff is of paramount importance, and schools have to be adequately prepared to address security threats and respond effectively.

Effective Response

Schools can respond effectively to threats by creating a safety plan, training staff and students, implementing security measures, and working with law enforcement agencies.



How is this problem being solved now?

Currently, gun detection in schools primarily depends on traditional security measures like:

- CCTV monitoring
- Metal detectors
- Manual inspections

These systems are **reactive** rather than **proactive**, often only activating after an incident has been reported.



Challenges in Ensuring School Safety

Multi-Layered Approach

Ensuring school safety requires a multi-layered approach that involves collaboration between schools, parents, law enforcement agencies, and other stakeholders.

Security Threats

Schools have to deal with a range of security threats, including gun violence, bullying, and cyber threats. These threats require a comprehensive and proactive approach to ensure safety.

Limitations of Traditional Measures

Traditional security measures, such as CCTV cameras, metal detectors, and security personnel, have limitations in ensuring school safety. These measures need to be complemented with technology-enabled solutions and proactive strategies.



Gun Detection Systems as a Solution

The Role of Gun Detection Systems

Enhancing School Safety

Gun detection systems can enhance school safety by detecting and alerting school officials to the presence of a gun in real-time, enabling them to respond quickly and effectively to security threats.

Advanced Technologies

Gun detection systems use advanced technologies such as acoustic sensors, video analytics, and artificial intelligence to detect and alert school officials to the presence of a gun in real-time, making schools safer for everyone.



Implementation of Gun Detection Systems

Considerations for Implementing a Gun Detection System

Implementing a gun detection system in schools requires careful planning and execution. School officials have to consider various factors such as cost, reliability, and privacy concerns.

Cost vs Benefit Analysis

Schools need to conduct a cost vs benefit analysis before investing in a gun detection system. They need to weigh the benefits of improved safety against the cost of the system and ongoing maintenance.

Rollout Strategy

Schools need to have a rollout strategy in place before implementing a gun detection system. This includes training staff, communicating with parents and students, and testing the system before going live.



Proposed End to End Solution

Data-Driven Detection

- Our model is trained on a dataset featuring police officers, civilians, and various firearms.
- This diverse collection is crucial for teaching our CNN to accurately spot civilians with guns amidst other individuals.

Model Training for Real-World Application

- By learning from images of both armed police and civilians, the CNN becomes adept at distinguishing between the two in the school's surveillance footage.
- The goal is for the model to identify when a gun is present with a civilian, a key indicator of potential danger.

From Detection to Response

- When our system identifies a civilian with a gun, it triggers an immediate alert.
- These alerts are sent directly to the school's security team and local law enforcement for a swift response.
- Every incident is recorded, feeding into our system's memory, which is used for future analysis and continuous improvement of the model.



Scope of the solution:

The solution is designed to cover all school areas monitored by CCTV. It aims to provide comprehensive coverage to ensure no potential threat goes undetected.

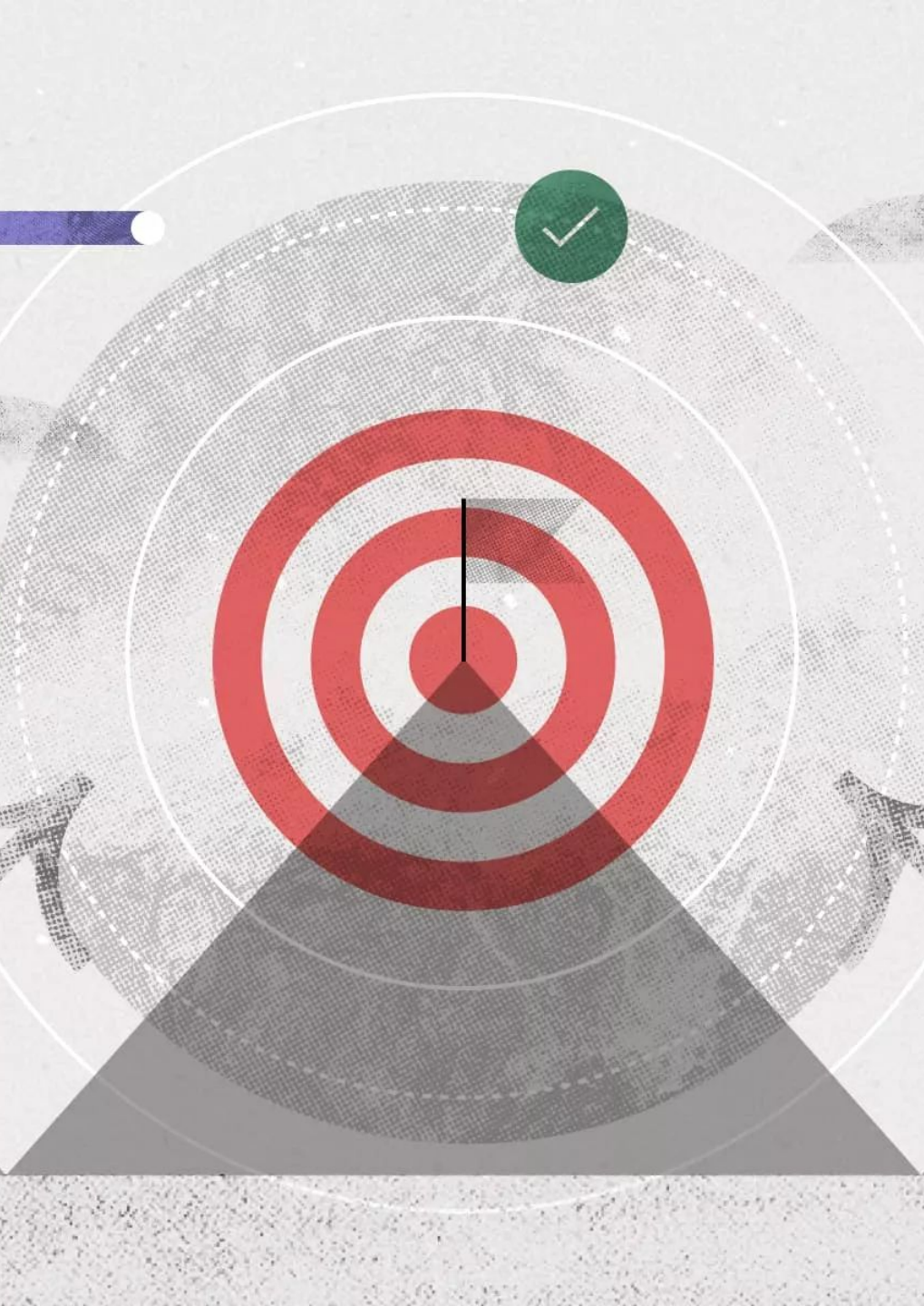
Key stakeholders and beneficiaries of the solution:

Stakeholders include school administrators, security personnel, parents, and law enforcement agencies. Beneficiaries are primarily students and staff, who will experience a safer educational environment.

Value of the solution:

The value lies in its potential to significantly reduce response times to armed threats, potentially saving lives. The utility for end-users is increased safety and peace of mind. The cost-benefit tradeoff considers the investment in technology versus the potential to prevent life-threatening incidents.





Success Metrics of this solution

Success metrics include the response time improvement, user satisfaction surveys among school staff and parents, and a measurable decrease in gun-related incidents in schools due to early detection.

Unintended Consequences and Privacy Concerns



Consequences: Privacy concerns, potential misidentification of non-threatening objects.

To mitigate these, Implement data encryption, access control, continuous training of the system, protocols for false positives.

Privacy and Security Concerns: Constant surveillance raises privacy issues.

Mitigation Strategies like Data encryption, restricted data access, compliance with privacy laws, regular audits are used.

Computer Vision



Application of Computer Vision (CV) in the Solution:

- **Real-time Analysis:** CV continuously monitors surveillance footage within schools.
- **Firearm Detection:** Identifies the presence of firearms in diverse settings.
- **Differentiation Capability:** Distinguishes between civilians and police, with or without firearms.

Why is CV required here?

- **Stops Danger Early:** Finds guns quickly before they can cause harm.
- **Better Than Old Ways:** Does more than metal detectors.

Choice of CV Model?

- **Multi-class Classification & Detection**

Classification Model

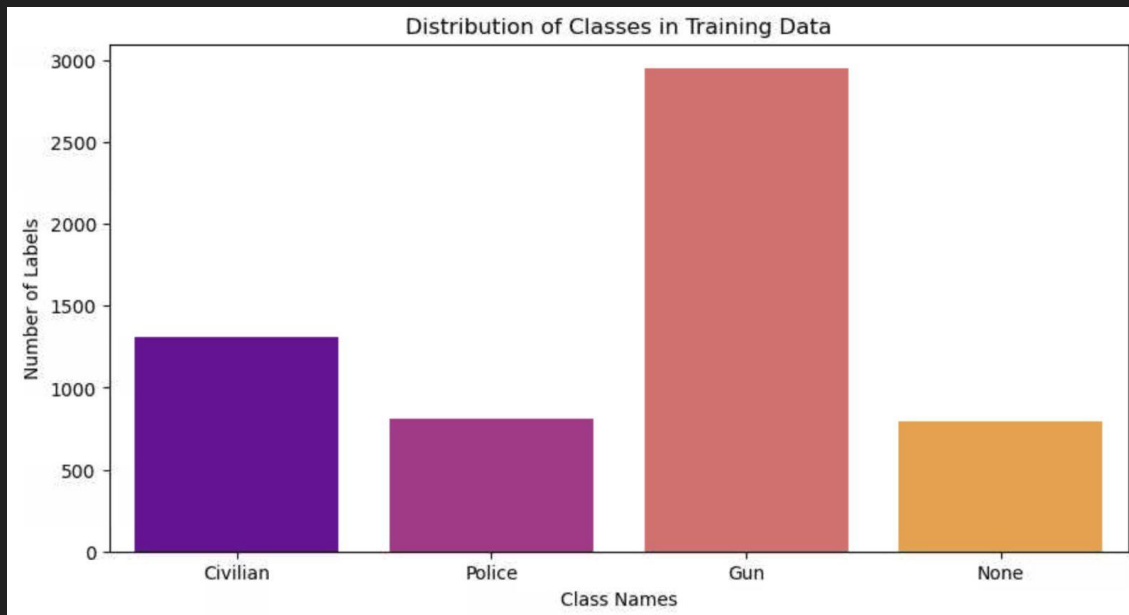
- Multi-Label Classification Problem
- Classes: Civilian, Police, Guns, None
- Guns - Kaggle Gun Detection Datasets
- People - COCO Datasets
- Police - GitHub CV Project Datasets

civilian	police	gun	none
0	1	0	0
0	1	0	0
0	0	1	0
0	0	1	0
0	0	1	0

Dataset Distribution

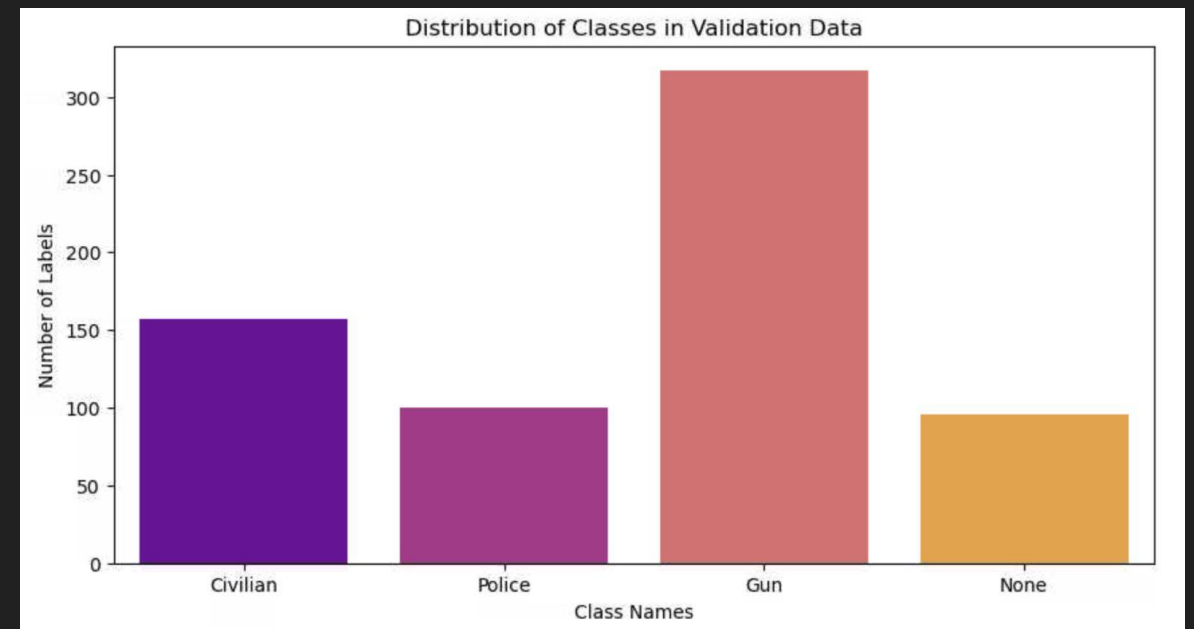
Total data in Training dataset: 5870

Civilian: 1308, Police: 814, Gun: 2951, None: 797



Total data in Validation dataset: 670

Civilian: 157, Police: 100, Gun: 317, None: 96



Dataset Sample Images



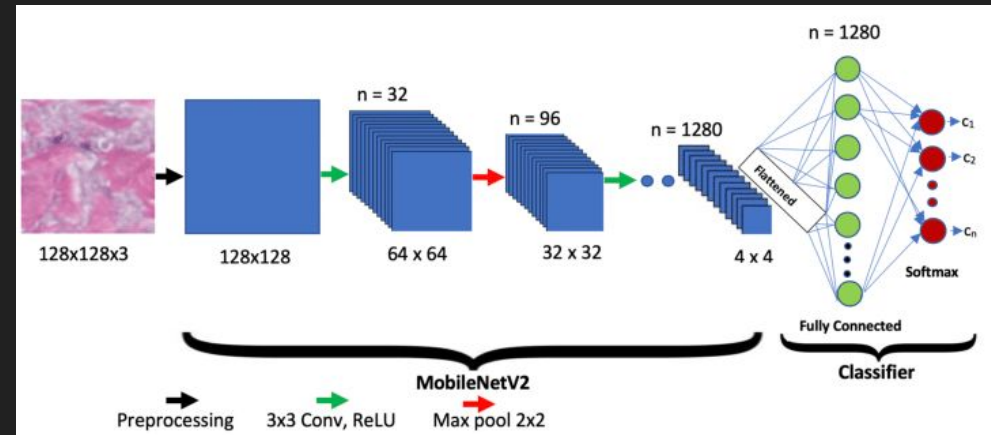
Built and Train the Model

Model

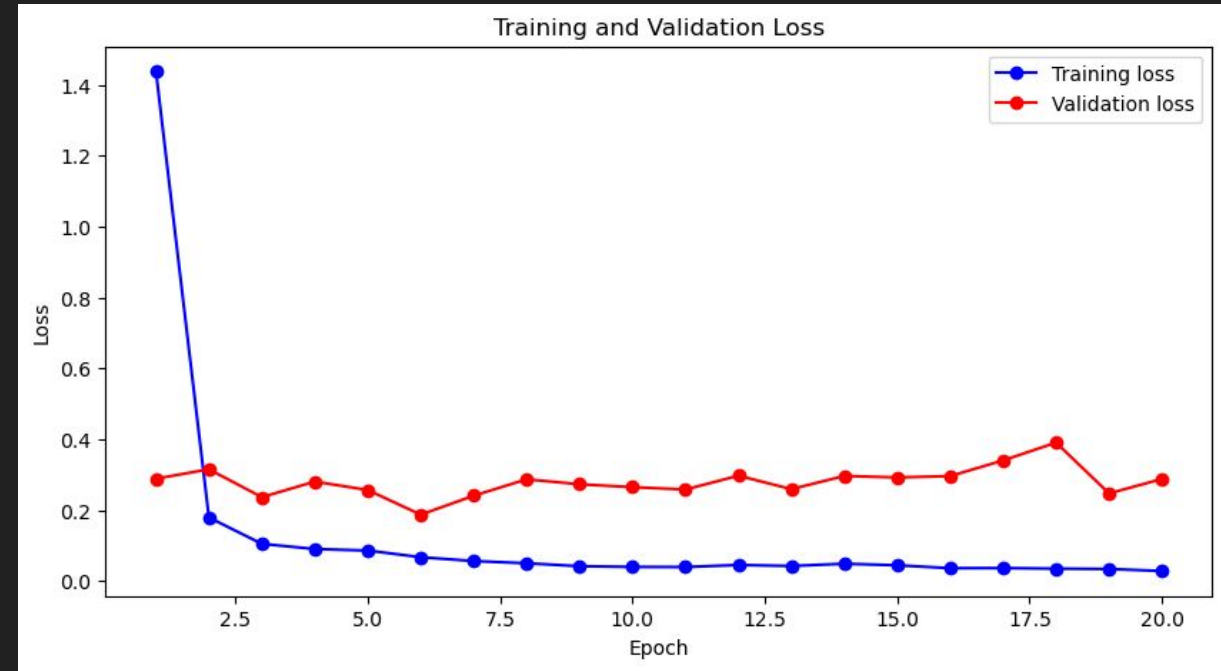
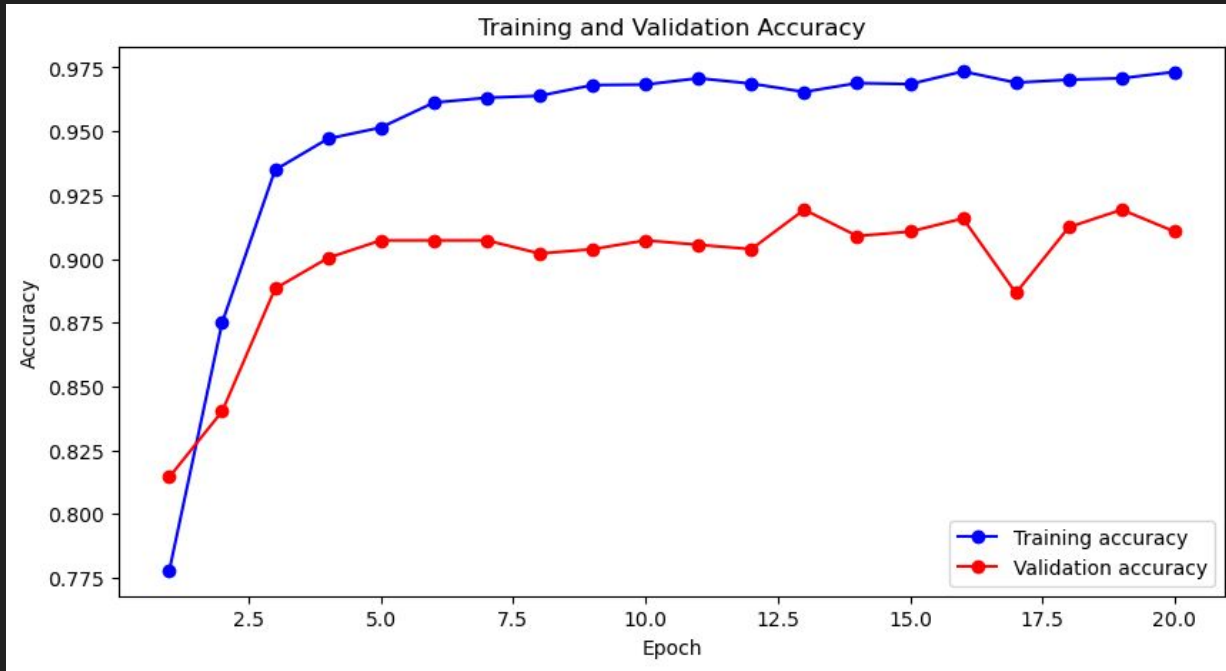
- MobileNetV2 pre trained with ImageNet
- Output Layer with 4 nodes and sigmoid activation
- Optimizer: Adam
- Loss = 'binary_crossentropy'
- Metrics = ['accuracy']

Train

- Batch size: 32
- Epochs: 20
- Learning rate=0.001



Training and Validation Accuracy and Loss



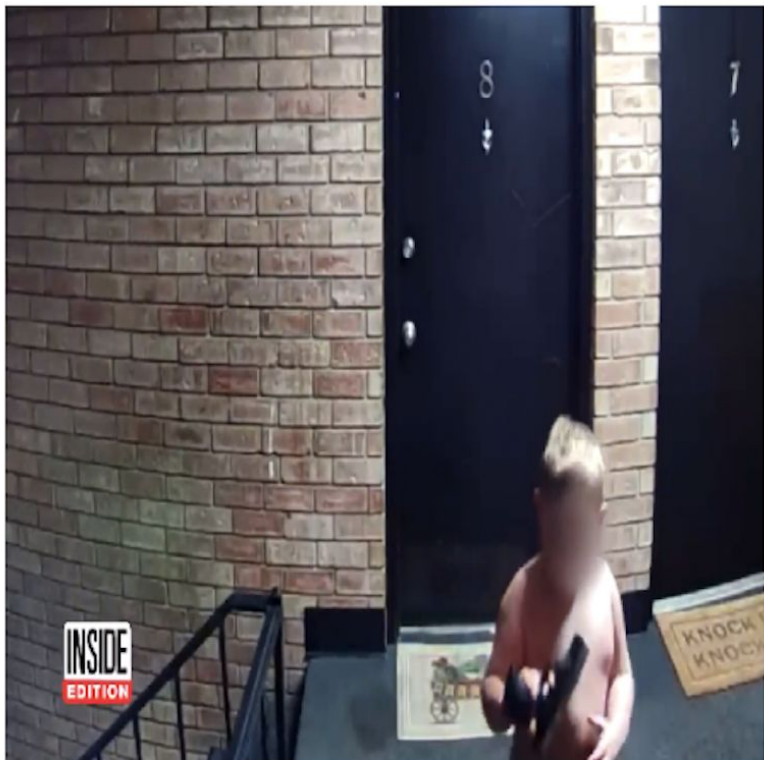
Model Evaluation

- We evaluated our model using 20 random images from the internet.
- This ensures the test set is different from the training and validation sets.
- This ensures our model will be able to perform well post deployment with new data.
- Model is very good in detecting Guns but can improve in classifying Civilians and Police.

Test Accuracy: 0.70				
	precision	recall	f1-score	support
Civilian	1.00	0.67	0.80	6
Police	1.00	0.67	0.80	3
Gun	1.00	1.00	1.00	8
None	1.00	1.00	1.00	2
micro avg	1.00	0.84	0.91	19
macro avg	1.00	0.83	0.90	19
weighted avg	1.00	0.84	0.91	19
samples avg	1.00	0.87	0.91	19

Successful Classifications by the CV Model

Actual: Civilian, Gun
Predicted: Civilian, Gun



Actual: Civilian, Gun
Predicted: Civilian, Gun



Actual: Police, Gun
Predicted: Police, Gun



Areas for Improvement in the CV Model

Actual: Police, Gun
Predicted: Gun



sky news 22:19 Rishi Sunak in the Conservative Party leadership contest, after going out in the first round

Actual: Civilian, Police, Gun
Predicted: Police, Gun



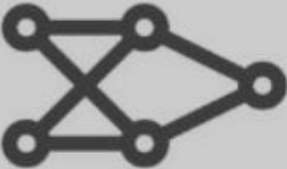
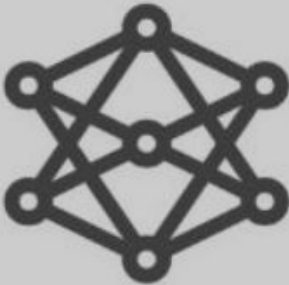
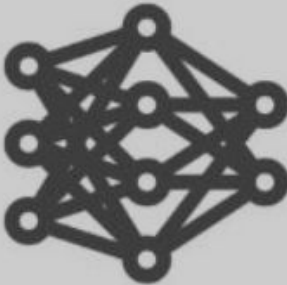
Actual: Civilian, Gun
Predicted: Gun



Object Detection

- Multiple Object Detection
- Re-trained a YOLOv5 pretrained model
- Classes to Detect: Guns and Persons
- We dropped the Police class due to lack of data.
- Person - COCO Dataset
- Guns - Kaggle Gun Detection Dataset with YOLO labels

Our Model - YOLOv5 Large

			
Small	Medium	Large	XLarge
YOLOv5s	YOLOv5m	YOLOv5l	YOLOv5x
14 MB _{FP16} 2.0 ms _{V100} 37.2 mAP _{COCO}	41 MB _{FP16} 2.7 ms _{V100} 44.5 mAP _{COCO}	90 MB _{FP16} 3.8 ms _{V100} 48.2 mAP _{COCO}	168 MB _{FP16} 6.1 ms _{V100} 50.4 mAP _{COCO}

Dataset Distribution

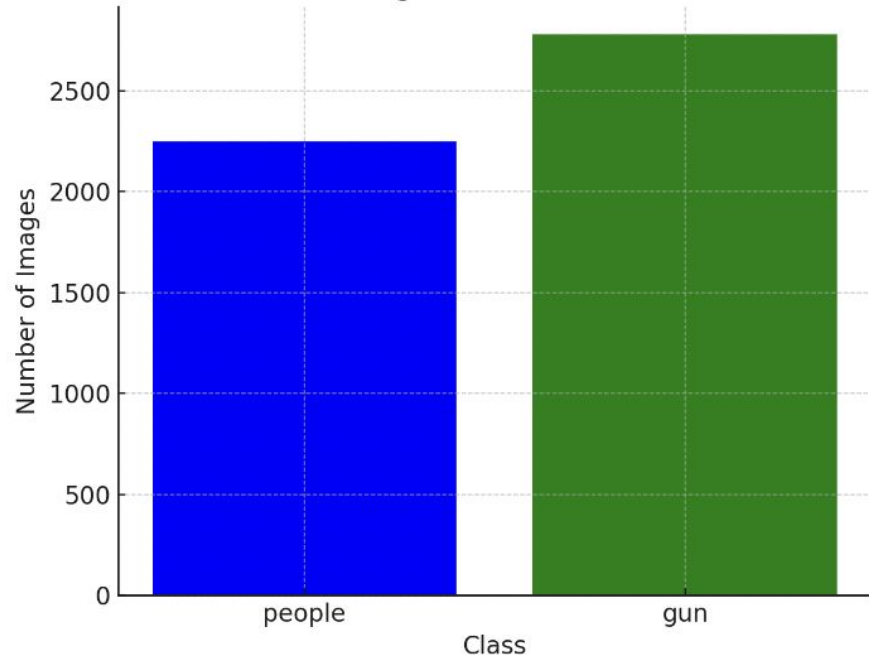
Total data in Training dataset: 5030

People: 2250, Gun: 2780

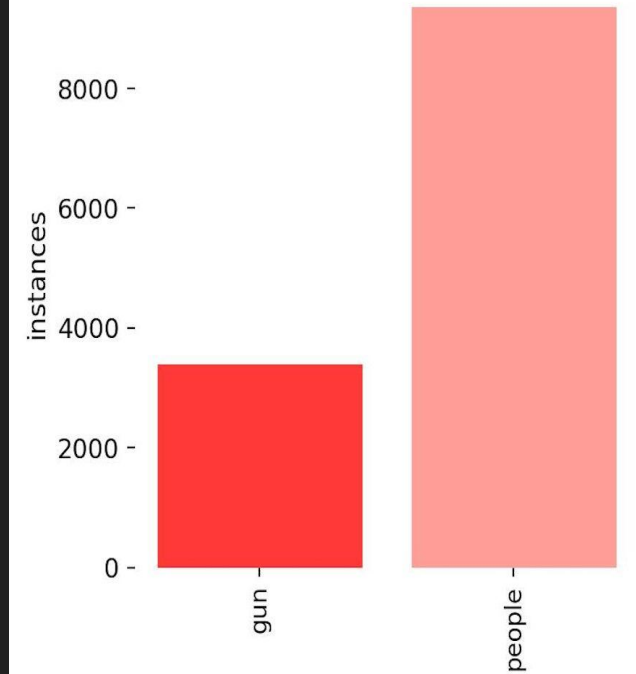
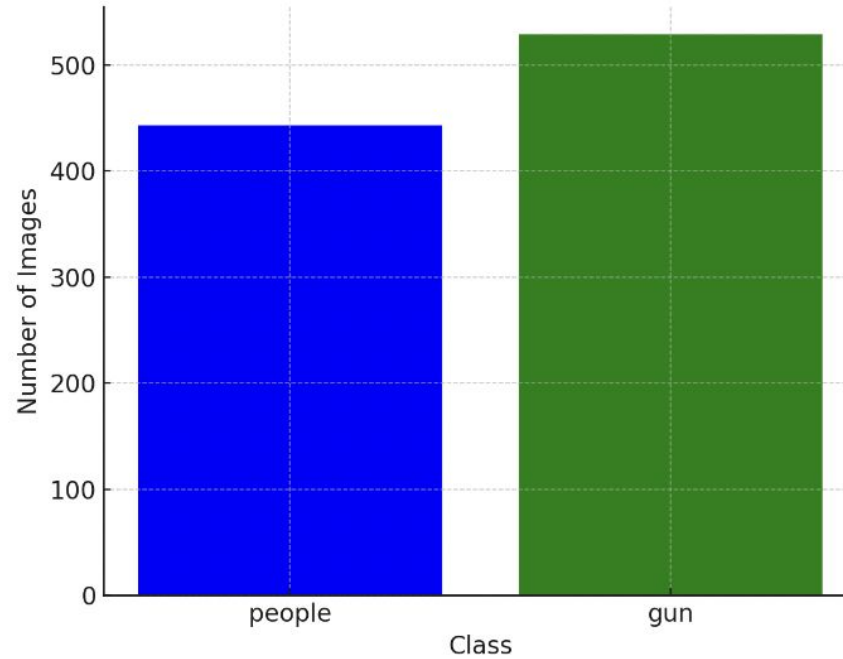
Total data in Validation dataset: 972

People: 443, Gun: 529

Training Data Distribution



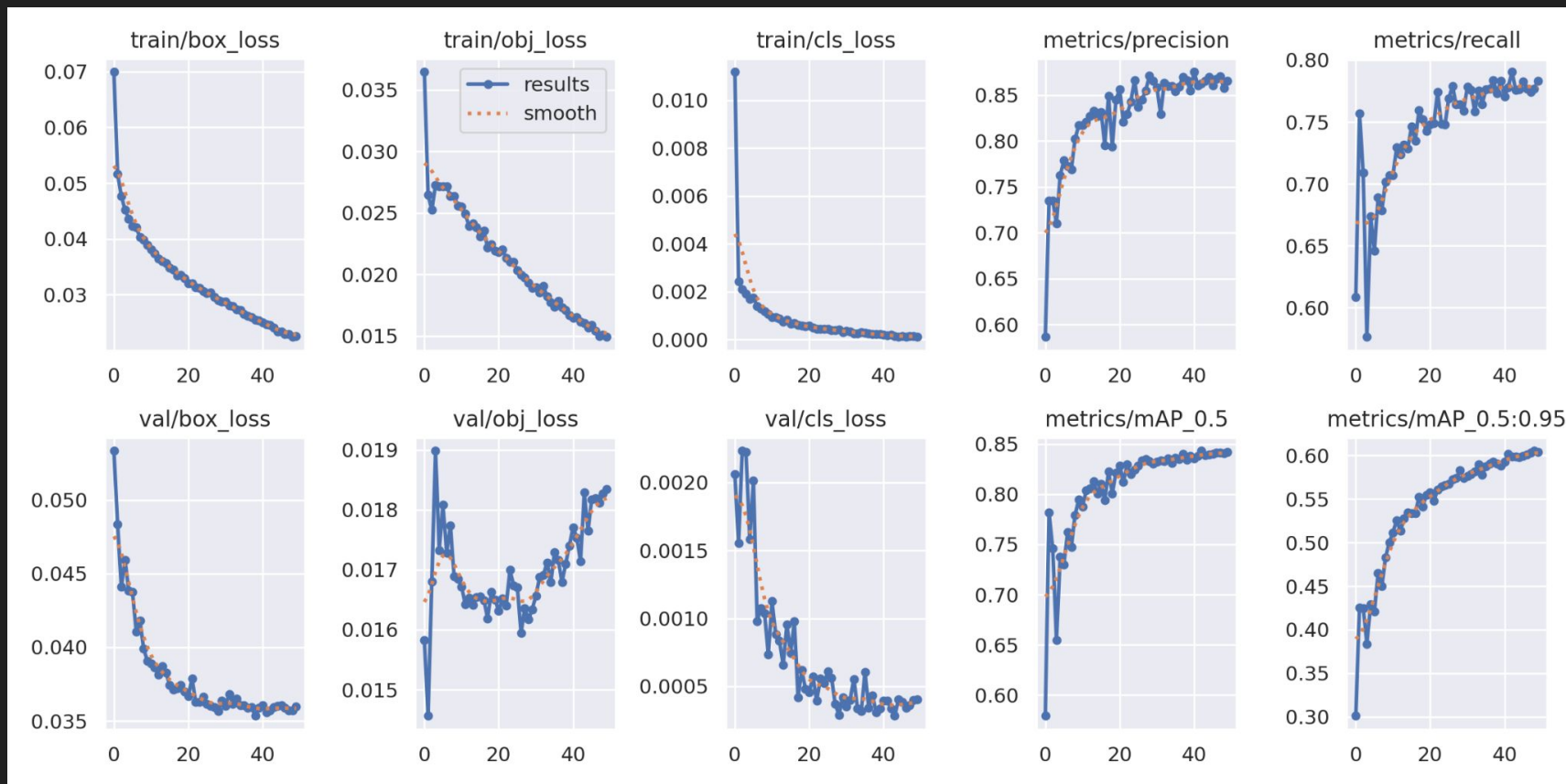
Validation Data Distribution



Dataset Sample Images



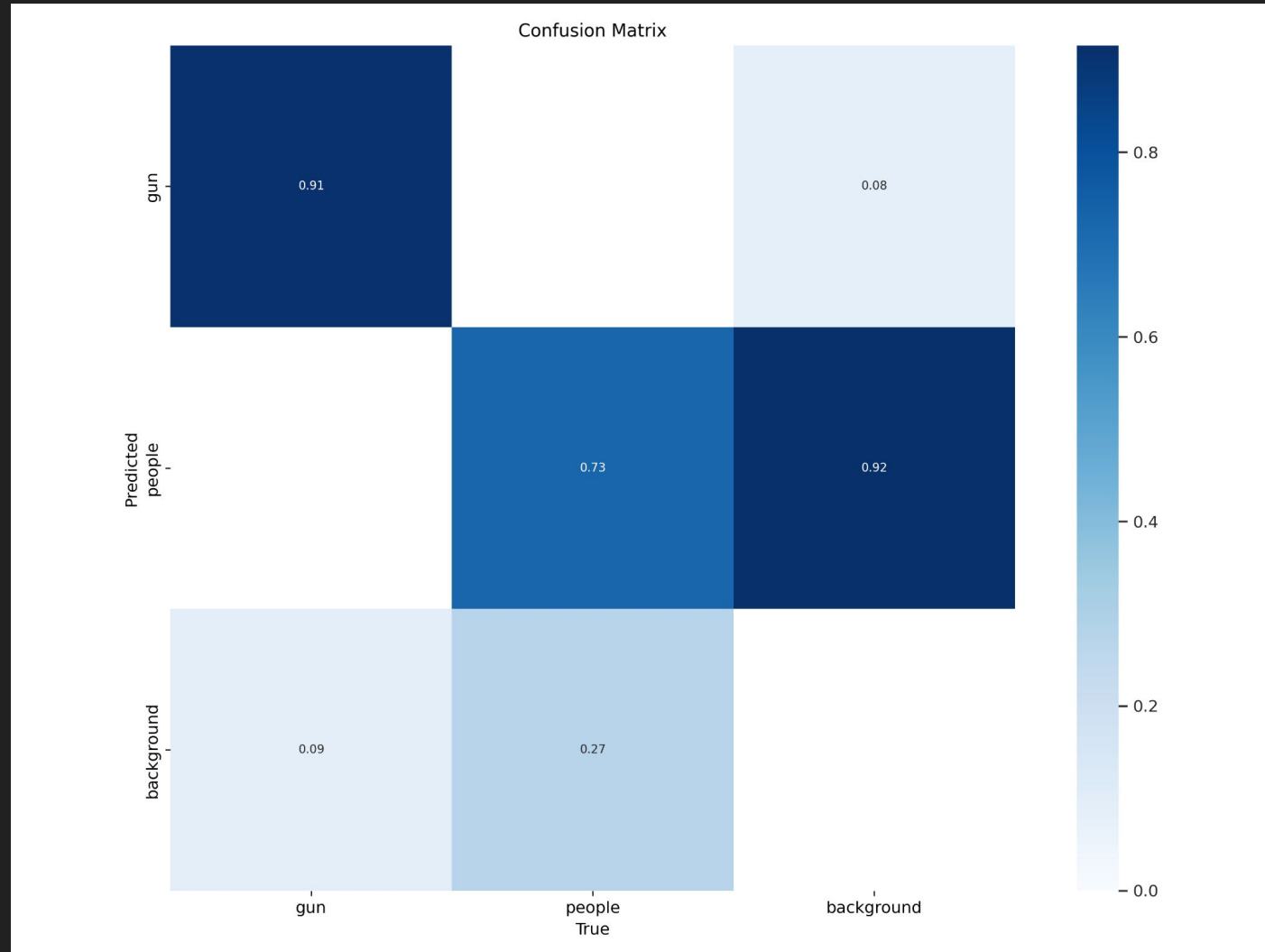
Training and Validation Metrics



Model summary: 267 layers, 46113663 parameters, 0 gradients, 107.7 GFLOPs

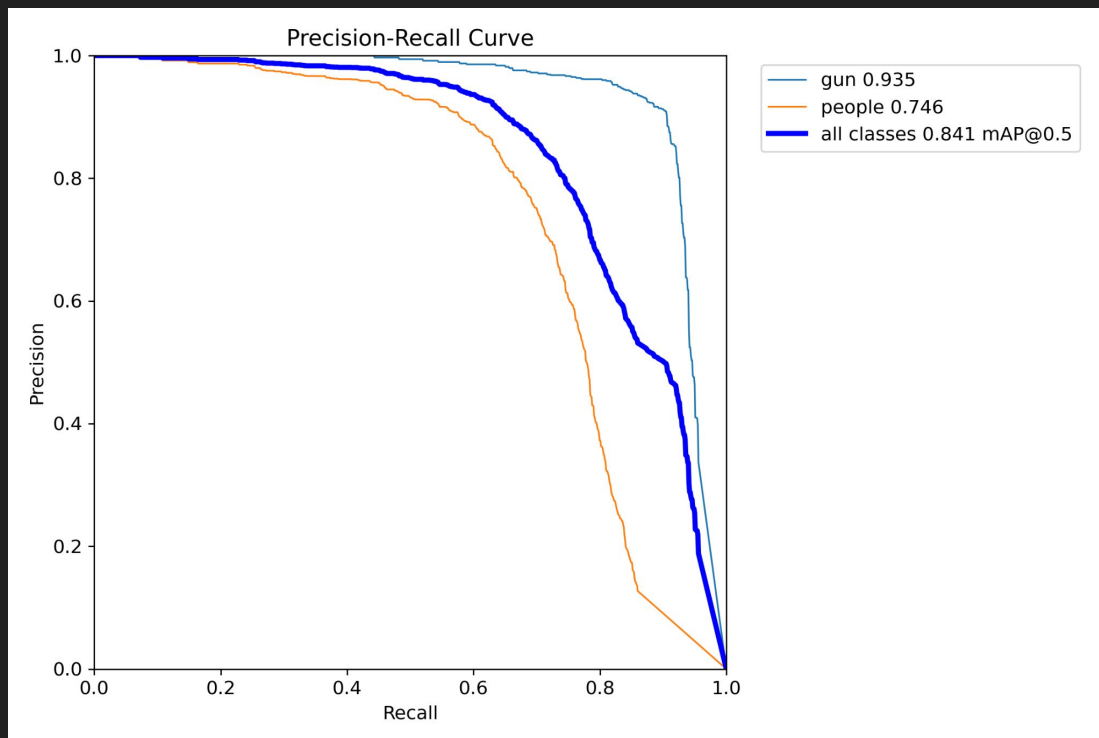
Class	Images	Instances	P	R	mAP50	mAP50-95:
all	972	2310	0.858	0.776	0.841	0.605
gun	972	652	0.923	0.879	0.935	0.705
people	972	1658	0.793	0.673	0.746	0.505

Model Confusion Matrix



- The matrix shows strong diagonal entries which indicate high true positive rates for both classes (guns and people).
- The model seems to have a small confusion between people and the background but performs well in distinguishing guns.

Precision - Recall



F1 Score

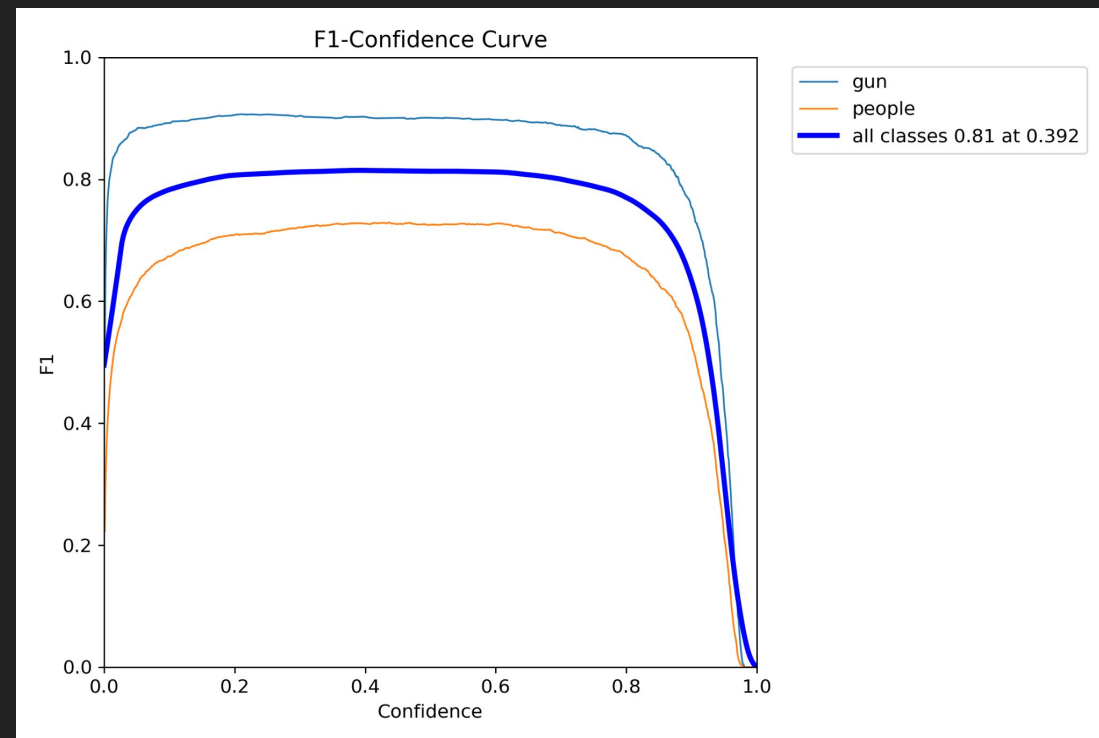
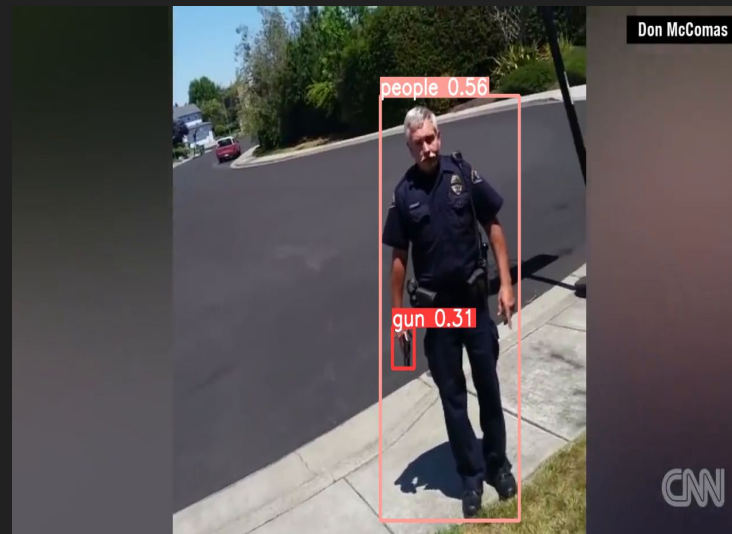
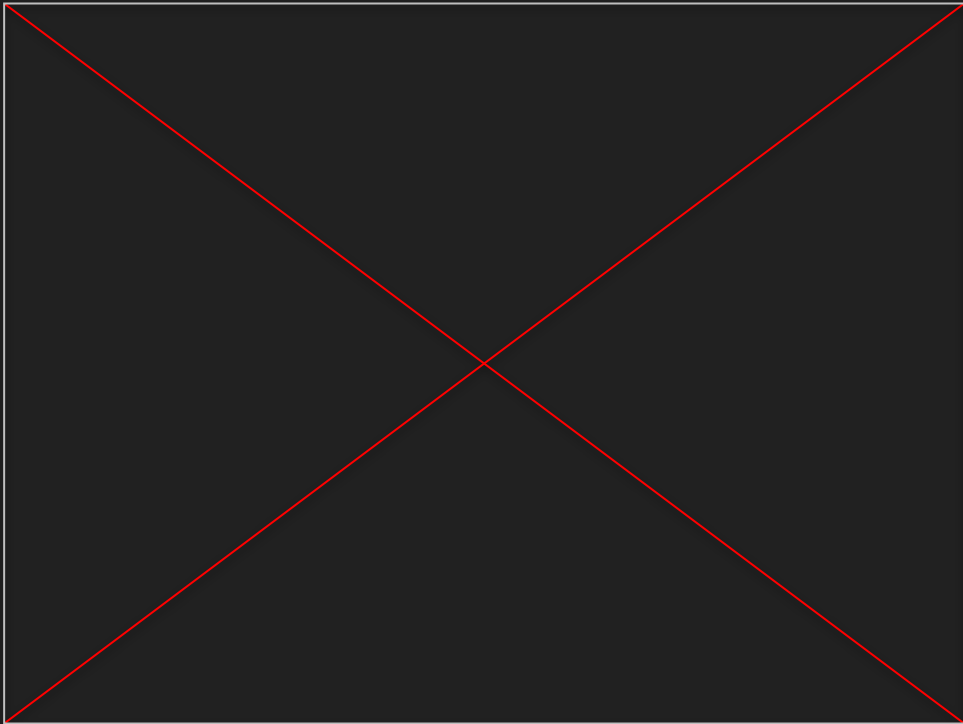


Image Detection



Video Detection



Validate the solution pre and post deployment

- Test the System
- Use diverse data to check accuracy.
- Identify false alarms and missed detections.
- Run Simulations
- Test the system in various school scenarios.
- Include emergency situations and busy times.
- Involve Stakeholders
- Get feedback from school staff, parents, and police.
- Ensure privacy and security rules are followed.

Monitor and Update After Deployment

- Continuous Monitoring
 - Keep track of the system's performance.
 - Set up alerts for quick response.
 - Regular Testing and Updates
 - Test the system regularly to maintain accuracy.
 - Update the system with new data.
 - Collect Feedback
 - Get feedback from stakeholders to identify issues.
 - Adjust the system based on feedback.
 - Ensure Privacy and Security
 - Use encryption to protect data.
 - Limit access to authorized personnel only.
-

Post-Deployment Plans & Outcome-Action Pairings

Post-Deployment Plans

- Continuous learning and model refinement would be necessary
- Regular testing with updated datasets to maintain high accuracy and reduce biases

Outcome-Action Pairings

- An alert sent to security personnel when a firearm is detected with a civilian
 - No action taken if a police officer is correctly identified with a gun.
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Bias Mitigation and Model Limitations

Bias Mitigation

- Expand and diversify the dataset
- Consider different scenarios

Model Limitations

- Include the challenge of detecting concealed weapons
- Address it through periodic model updates and possibly integrating other data sources or sensors.

Scaling Gun Detection Systems

Expand Scope

- Include more school areas: classrooms, hallways, etc.
- Extend to universities and public spaces.

Diverse Domains

- Apply in offices, airports, and government buildings.
- Customize for each place's needs.

Various Scenarios

- Use during school, after-school events, and emergencies.
- Quick response in urgent situations.

Scale Technology

- Add features like facial recognition.
- Use cloud technology for easy management.

Collaborate

- Work with school staff, parents, and local police.
- Address privacy and security concerns.

Continuous Improvement

- Keep the system updated and accurate.
- Test regularly to maintain effectiveness.

Conclusion

- **Current model:**

Classification model: excels in gun detection, accurately distinguishing between police and civilian

Detection model: detects both individuals and firearms.

- **Future work:**

expanding the dataset to include police-related annotations and refining the models to boost performance

- **Ideal outcome:**

Gun detection systems offer real-time detection, enabling swift responses to potential violence, unlike reactive measures currently employed by schools.

Task Ownership and Contribution

Name	Tasks
Kashyap Mehta	Classification and Object Detection Model Building, Training & Testing, Deck Review
Lin-Hsuan Chen	Dataset Collection, Classification Model Testing, CV Model Slides
Pin-Lin Lai	Test Dataset, Evaluation Metrics, End to end solution Slides
Meshvee Kanani	Dataset and Annotation preparation, Multi Label Classification Research, End to end solution, Applications of Computer Vision Slides
Narayanan TK	Data collection, Detection Research, Presentation outline creation, Slides on project introduction to slides explaining how it is being solved currently

References

- [YOLOv5 for Object Detection](#)
- [MobileNetv2](#)
- [Multi-Label Classification](#)
- [YouTube - Keras Implementation](#)
- [Object Detection Model & Results](#)
- [CIS515 Project.ipynb](#)
- [Gun Detection with YOLO Labels](#)
- [Weapon Detection Dataset](#)
- [Handgun Detection for Classification](#)
- [Police Dataset](#)
- [Civilians Dataset](#)